

UNIT 2: Programming and Interrupt structure

1. What is the primary function of the instruction set in the 8086 microprocessor?

- A) To manage memory
- B) To perform calculations
- C) To provide a list of operations the CPU can execute
- D) To control input/output devices

Answer: C) To provide a list of operations the CPU can execute

Justification: The instruction set defines the operations that the microprocessor can perform, including arithmetic, logic, and control operations.

2. Which of the following is an example of a data transfer instruction in 8086?

- A) ADD
- B) MOV
- C) CMP
- D) JMP

Answer: B) MOV

Justification: The MOV instruction is used to transfer data from one location to another.

3. Which addressing mode uses a register to hold the operand?

- A) Immediate addressing
- B) Direct addressing
- C) Register addressing
- D) Indirect addressing

Answer: C) Register addressing

Justification: In register addressing mode, the operand is located in a register.

4. In which addressing mode is the operand specified directly in the instruction?

- A) Direct addressing
- B) Indirect addressing
- C) Indexed addressing
- D) Register indirect addressing

Answer: A) Direct addressing

Justification: Direct addressing mode specifies the operand directly within the instruction.

5. What is the purpose of the assembler directive `ORG`?

- A) To define a variable
- B) To specify the starting address of the program
- C) To allocate memory space
- D) To comment code

Answer: B) To specify the starting address of the program

Justification: The `ORG` directive is used to indicate the origin or starting address for the subsequent code or data.

6. Which directive is used to define a variable in assembly language?

- A) DB
- B) DW
- C) EQU
- D) END

Answer: A) DB

Justification: The `DB` directive is used to declare a byte-sized variable in memory.

7. How does the `INT` instruction work in the 8086 microprocessor?

- A) It halts the processor
- B) It performs a software interrupt
- C) It clears a register
- D) It manages memory

Answer: B) It performs a software interrupt

Justification: The `INT` instruction triggers a software interrupt, allowing the program to call an interrupt service routine.

8. What type of interrupt is a hardware interrupt?

- A) INT
- B) INTR
- C) INT 21h
- D) INT 10h

Answer: B) INTR

Justification: INTR (interrupt request) is a hardware interrupt signal that can interrupt the CPU to request service from an interrupt handler.

9. Which of the following is NOT an addressing mode supported by the 8086?

- A) Register addressing
- B) Indexed addressing
- C) Immediate addressing
- D) Floating-point addressing

Answer: D) Floating-point addressing

Justification: The 8086 supports various addressing modes but does not have a distinct floating-point addressing mode.

10. In which addressing mode is the effective address calculated by adding a constant value to the contents of a register?

- A) Direct addressing
- B) Indirect addressing
- C) Indexed addressing
- D) Base addressing

Answer: C) Indexed addressing

Justification: Indexed addressing involves adding an offset (constant) to a base address held in a register.

11. What is the function of the assembler directive `EQU`?

- A) To reserve memory
- B) To define a constant
- C) To end the program
- D) To define a procedure

Answer: B) To define a constant

Justification: The `EQU` directive is used to define a constant value that can be used throughout the program.

12. What type of instruction is `CALL` in 8086 assembly language?

- A) Data transfer instruction
- B) Arithmetic instruction
- C) Control transfer instruction
- D) Logic instruction

Answer: C) Control transfer instruction

Justification: The `CALL` instruction is used to transfer control to a subroutine or procedure.

13. Which of the following interrupt types allows the processor to respond to external events?

- A) Software interrupt
- B) Timer interrupt
- C) Hardware interrupt
- D) I/O interrupt

Answer: C) Hardware interrupt

Justification: Hardware interrupts are generated by external devices to signal the CPU for processing.

14. What does the assembler directive `END` signify?

- A) The end of a loop
- B) The end of a procedure
- C) The end of the program
- D) The end of a data segment

Answer: C) The end of the program

Justification: The `END` directive indicates the termination of the assembly language program.

15. In the addressing mode "indirect addressing," where is the operand located?

- A) In a register
- B) Directly in the instruction
- C) At the address contained in a register
- D) At a fixed address

Answer: C) At the address contained in a register

Justification: Indirect addressing means the operand's address is stored in a register, and the CPU uses that address to access the data.

16. Which instruction is used to return from a procedure in 8086 assembly?

- A) RET
- B) RETURN
- C) EXIT
- D) HALT

Answer: A) RET

Justification: The `RET` instruction is specifically used to return control from a subroutine to the calling code.

17. Which of the following is an example of an immediate addressing mode?

- A) MOV AX, 5
- B) MOV AX, BX
- C) MOV AX, [2000h]
- D) MOV AX, [BX]

Answer: A) MOV AX, 5

Justification: The instruction `MOV AX, 5` uses immediate addressing, where the value 5 is specified directly in the instruction.

18. The `PUSH` instruction is used for which of the following?

- A) To transfer data
- B) To store a value on the stack
- C) To clear a register
- D) To call a subroutine

Answer: B) To store a value on the stack

Justification: The `PUSH` instruction places a value onto the stack, managing function calls and local variables.

19. Which of the following interrupts is triggered by a divide error?

- A) INT 00h
- B) INT 01h
- C) INT 02h
- D) INT 03h

Answer: A) INT 00h

Justification: The divide error is handled by the interrupt vector INT 00h in the 8086 architecture.

20. In the 8086, what is the purpose of the `IRET` instruction?

- A) To initiate an interrupt
- B) To return from an interrupt service routine
- C) To terminate a program
- D) To enable interrupts

Answer: B) To return from an interrupt service routine

Justification: The `IRET` instruction is used to return from an interrupt service routine and restore the previous state of the processor.